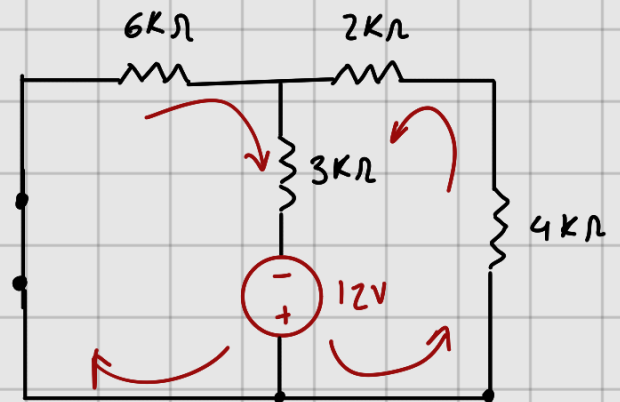
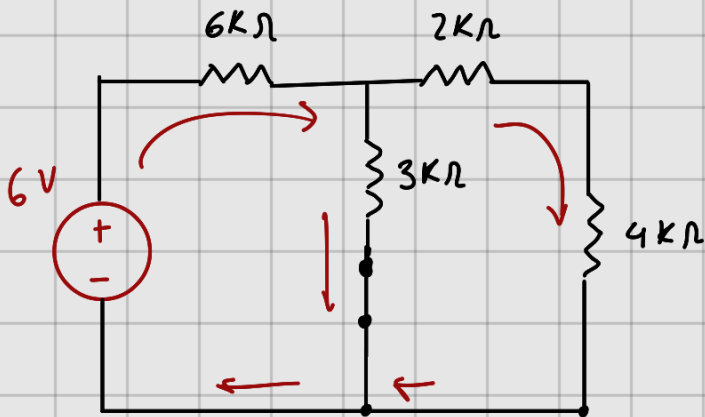
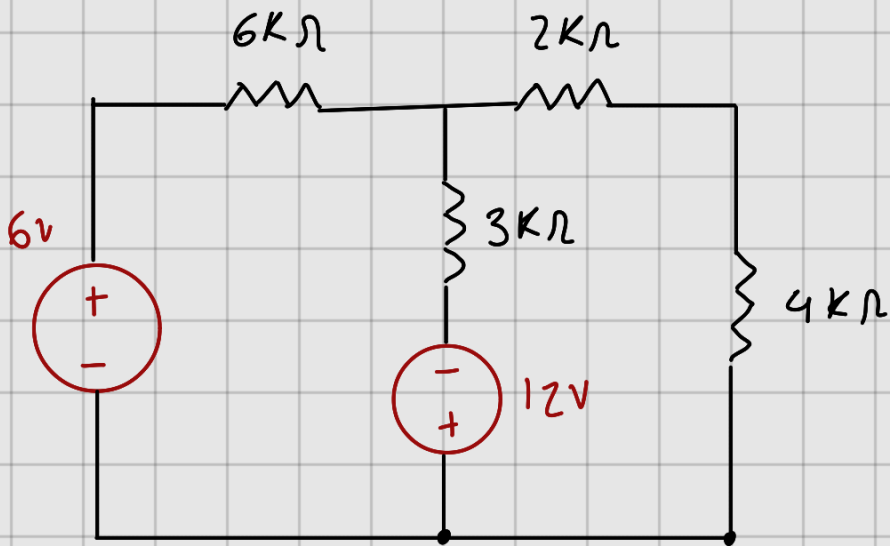
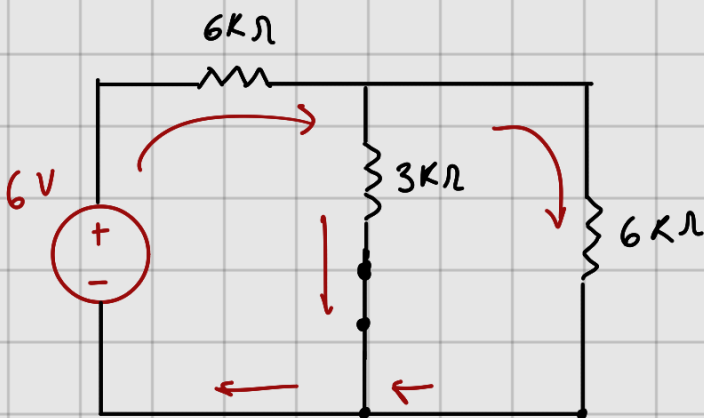
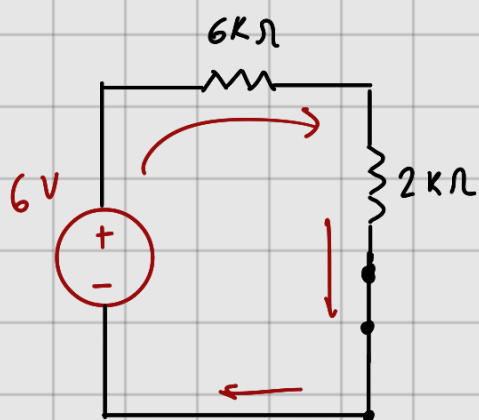




$$V_{4k} = V_T \frac{R_{4k}}{R_T}$$



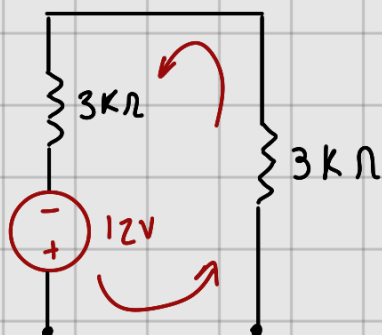
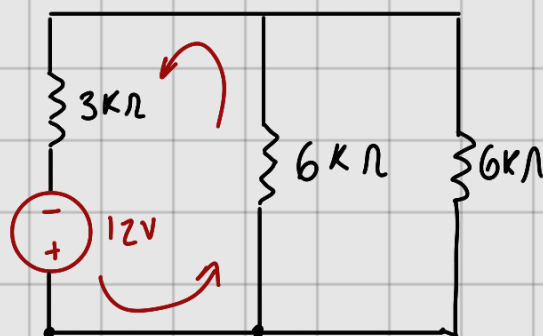
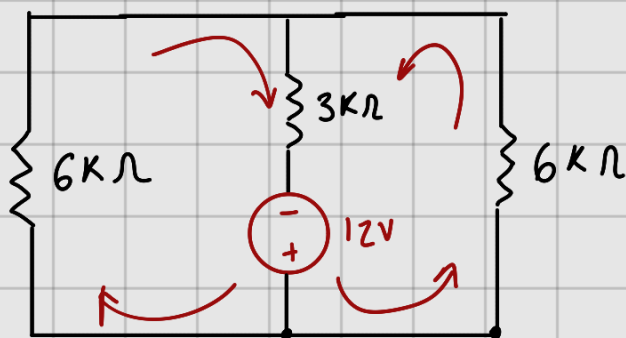
$$R_A = \frac{1}{\frac{1}{3k\Omega} + \frac{1}{6k\Omega}} = 2k$$



$$V_{2k} = 1V$$

$$R_T = 6k\Omega + 2k\Omega = 8k\Omega$$

$$I_T = \frac{6V}{8k\Omega} = 0.75mA$$



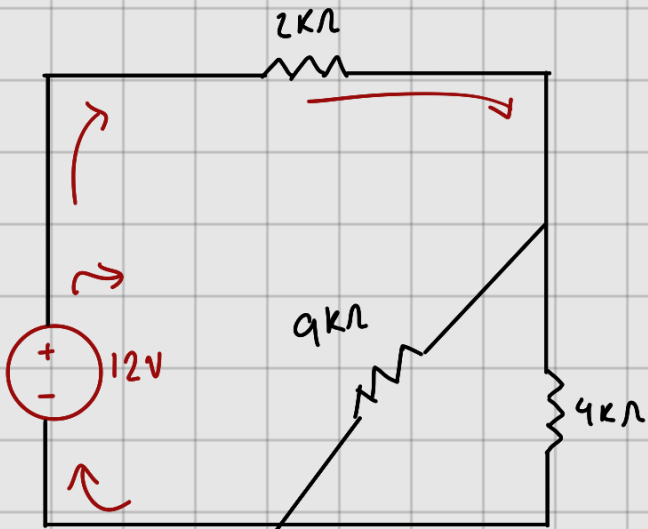
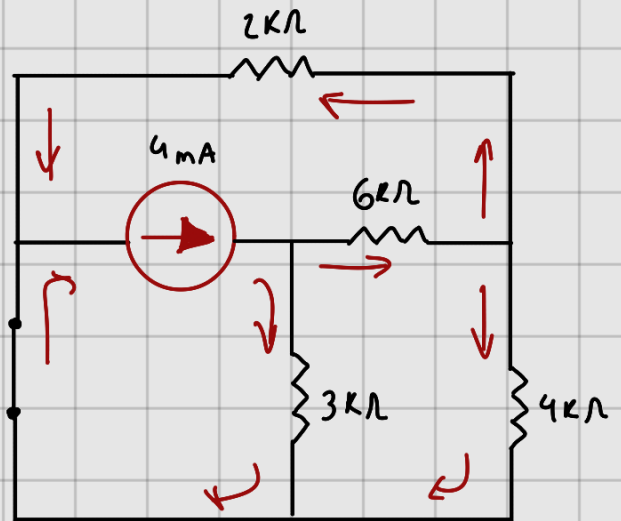
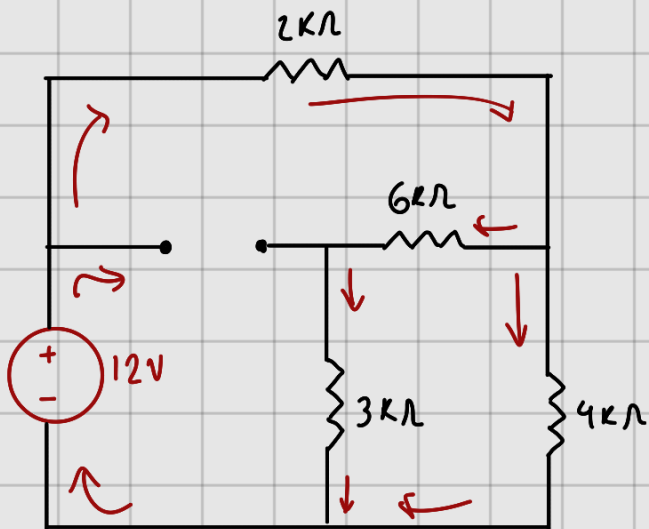
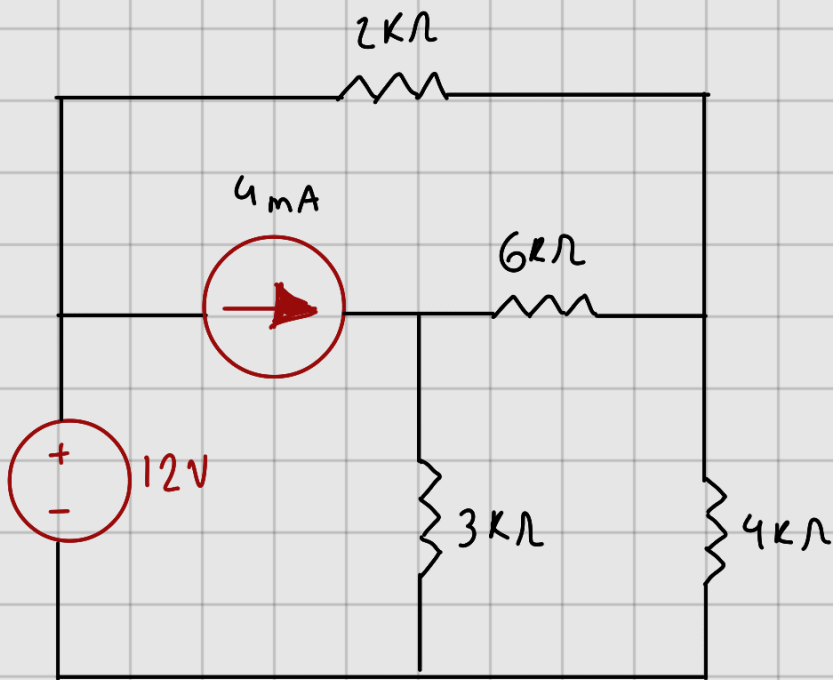
$$R_T = 6k\Omega$$

$$I_T = 2mA$$

$$V_{4k\Omega} = 4k\Omega (1mA) = 4V$$

$$V_{2k} = 4V - 1V = 3V$$

$$\underline{V_{4k} = 3V}$$

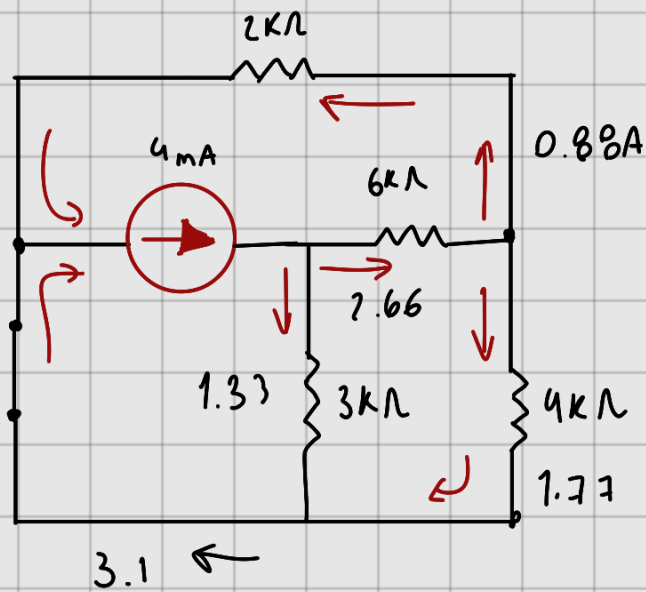


$$R_A = \frac{1}{\frac{1}{9k\Omega} + \frac{1}{4k\Omega}} = 2.769k\Omega$$

$$R_T = 4.769k\Omega$$

$$I_T = \frac{12V}{4.769k\Omega} = 2.51mA$$

$$V_{2k\Omega} = (2.51mA)(2k\Omega) = 5.02V$$



$$V_{2k} = 0.88A(2k\Omega)$$

$$V_{2k} = 1.776V$$

$$V_{2k\Omega} = 5.02V - 1.776V = 3.244$$

$$\underline{V_{2k\Omega} = 3.244V}$$